

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
%matplotlib inline
```

```
In [2]: df = pd.read_csv("train.csv")
```

```
In [3]: df.shape
```

```
Out[3]: (891, 12)
```

```
In [4]: df.head()
```

```
Out[4]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450

```
In [5]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

```

In [6]: `df.columns`

Out[6]: Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp', 'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'], dtype='object')

In [7]: `df.describe()`

Out[7]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000

In [8]: `df.isna().sum()`

```
Out[8]: PassengerId      0
        Survived        0
        Pclass          0
        Name            0
        Sex             0
        Age            177
        SibSp           0
        Parch           0
        Ticket          0
        Fare            0
        Cabin          687
        Embarked        2
        dtype: int64
```

```
In [9]: df["Age"] = df["Age"].fillna(df["Age"].mean())
```

```
In [10]: df.isna().sum()
```

```
Out[10]: PassengerId      0
        Survived        0
        Pclass          0
        Name            0
        Sex             0
        Age             0
        SibSp           0
        Parch           0
        Ticket          0
        Fare            0
        Cabin          687
        Embarked        2
        dtype: int64
```

```
In [11]: df["Name"]
```

```
Out[11]: 0                                Braund, Mr. Owen Harris
        1      Cumings, Mrs. John Bradley (Florence Briggs Th...
        2                                Heikkinen, Miss. Laina
        3      Futrelle, Mrs. Jacques Heath (Lily May Peel)
        4                                Allen, Mr. William Henry
        ...
        886                                Montvila, Rev. Juozas
        887                                Graham, Miss. Margaret Edith
        888      Johnston, Miss. Catherine Helen "Carrie"
        889                                Behr, Mr. Karl Howell
        890                                Dooley, Mr. Patrick
        Name: Name, Length: 891, dtype: object
```

```
In [12]: df["Sex"].value_counts()
```

```
Out[12]: Sex
        male      577
        female    314
        Name: count, dtype: int64
```

```
In [13]: df["Ticket"].value_counts()
```

```
Out[13]: Ticket
347082      7
CA. 2343    7
1601        7
3101295     6
CA 2144     6
..
9234        1
19988       1
2693        1
PC 17612    1
370376      1
Name: count, Length: 681, dtype: int64
```

```
In [14]: df["Cabin"].value_counts()
```

```
Out[14]: Cabin
B96 B98      4
G6           4
C23 C25 C27  4
C22 C26      3
F33          3
..
E34          1
C7           1
C54          1
E36          1
C148         1
Name: count, Length: 147, dtype: int64
```

```
In [15]: df["Embarked"].value_counts()
```

```
Out[15]: Embarked
S      644
C      168
Q       77
Name: count, dtype: int64
```

```
In [17]: def fun1(value):
         if (value == "male"):
             return 1
         else:
             return 0
```

```
In [18]: def fun2(value):
         if (value == 'S'):
             return 0
         elif (value == 'C'):
             return 1
         elif (value == 'Q'):
             return 2
         else:
             return 0
```

```
In [19]: df["Sex"] = df["Sex"].apply(fun1)
```

```
In [20]: df["Embarked"] = df["Embarked"].apply(fun2)
```

```
In [21]: df.isna().sum()
```

```
Out[21]: PassengerId      0
Survived      0
Pclass      0
Name      0
Sex      0
Age      0
SibSp      0
Parch      0
Ticket      0
Fare      0
Cabin      687
Embarked      0
dtype: int64
```

```
In [22]: df = df.drop("Cabin", axis=1)
```

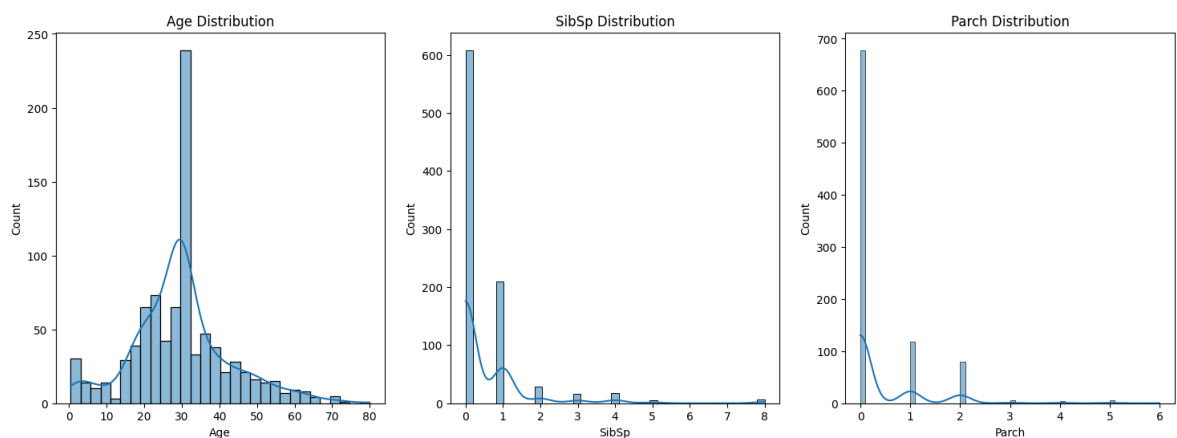
```
In [23]: df.shape
```

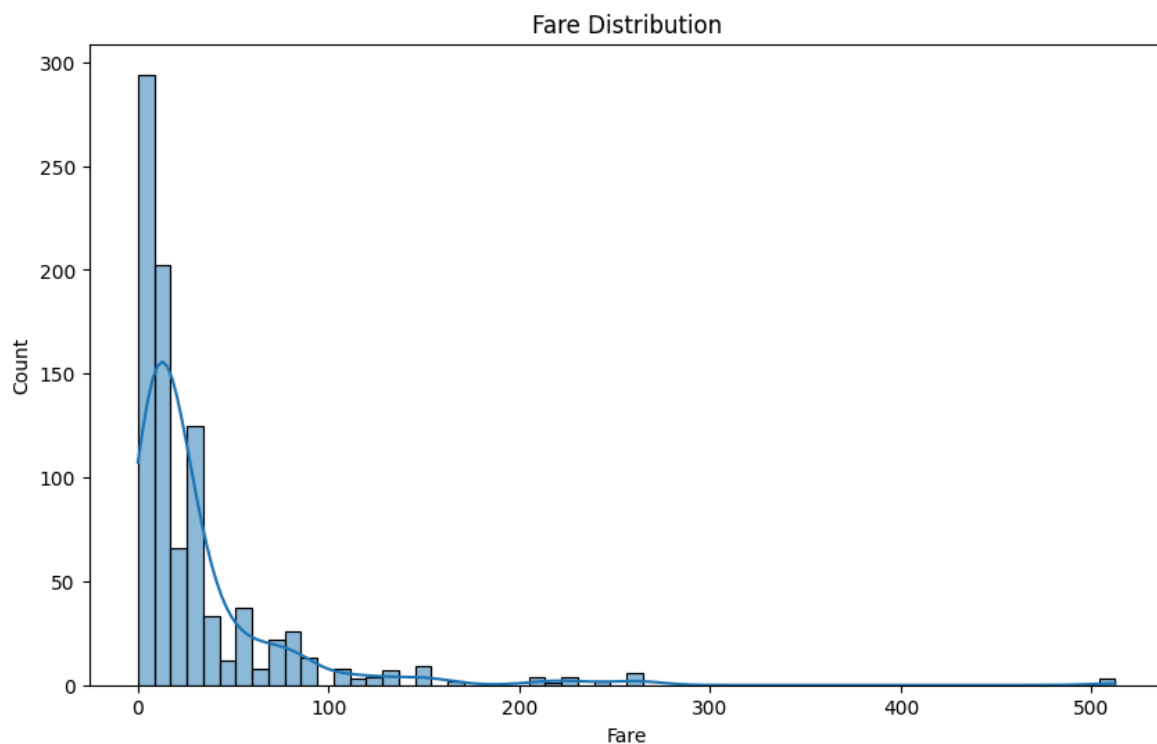
```
Out[23]: (891, 11)
```

```
In [24]: df.shape
```

```
Out[24]: (891, 11)
```

```
In [25]: #Set up the figure and axes
fig, axes = plt.subplots(1, 3, figsize=(18, 6))
# Age Distribution
sns.histplot(data=df, x='Age', kde=True, ax=axes[0])
axes[0].set_title('Age Distribution')
# SibSp Distribution
sns.histplot(data=df, x='SibSp', kde=True, ax=axes[1])
axes[1].set_title('SibSp Distribution')
# Parch Distribution
sns.histplot(data=df, x='Parch', kde=True, ax=axes[2])
axes[2].set_title('Parch Distribution')
#plt.tight_layout()
#plt.show()
# Fare Distribution
plt.figure(figsize=(10, 6))
sns.histplot(data=df, x='Fare', kde=True)
plt.title('Fare Distribution')
plt.show()
```

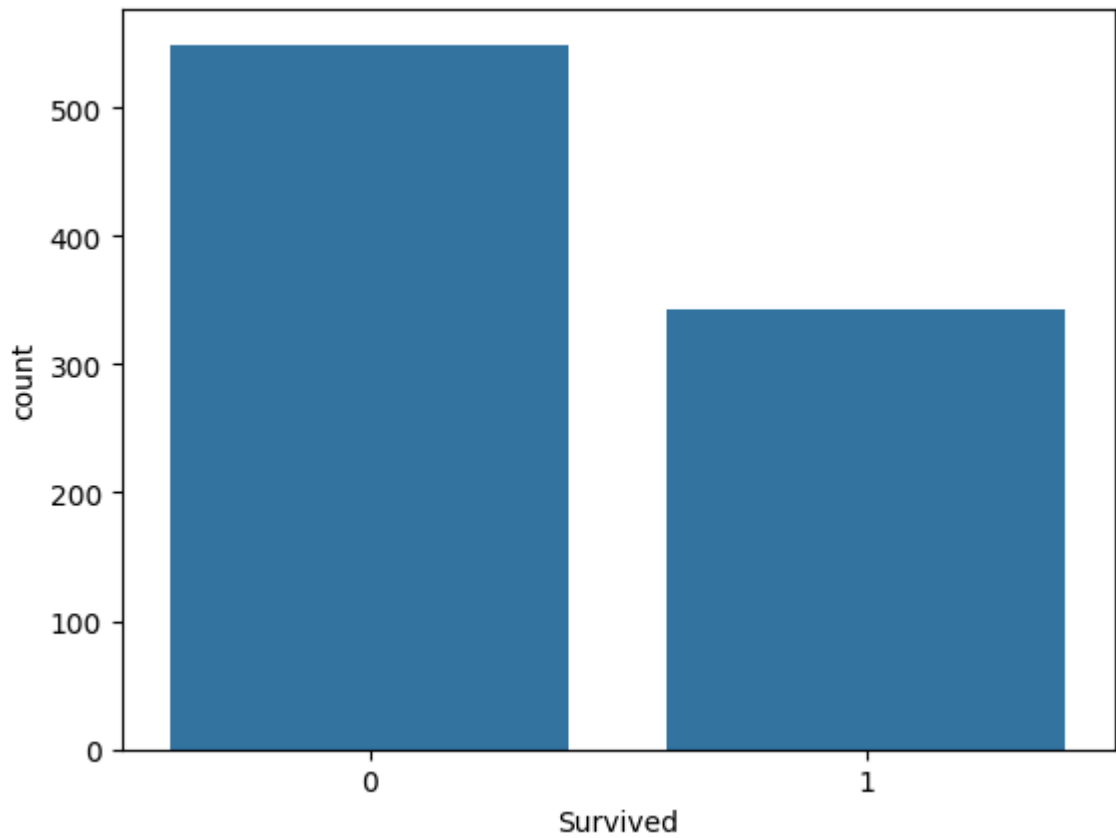




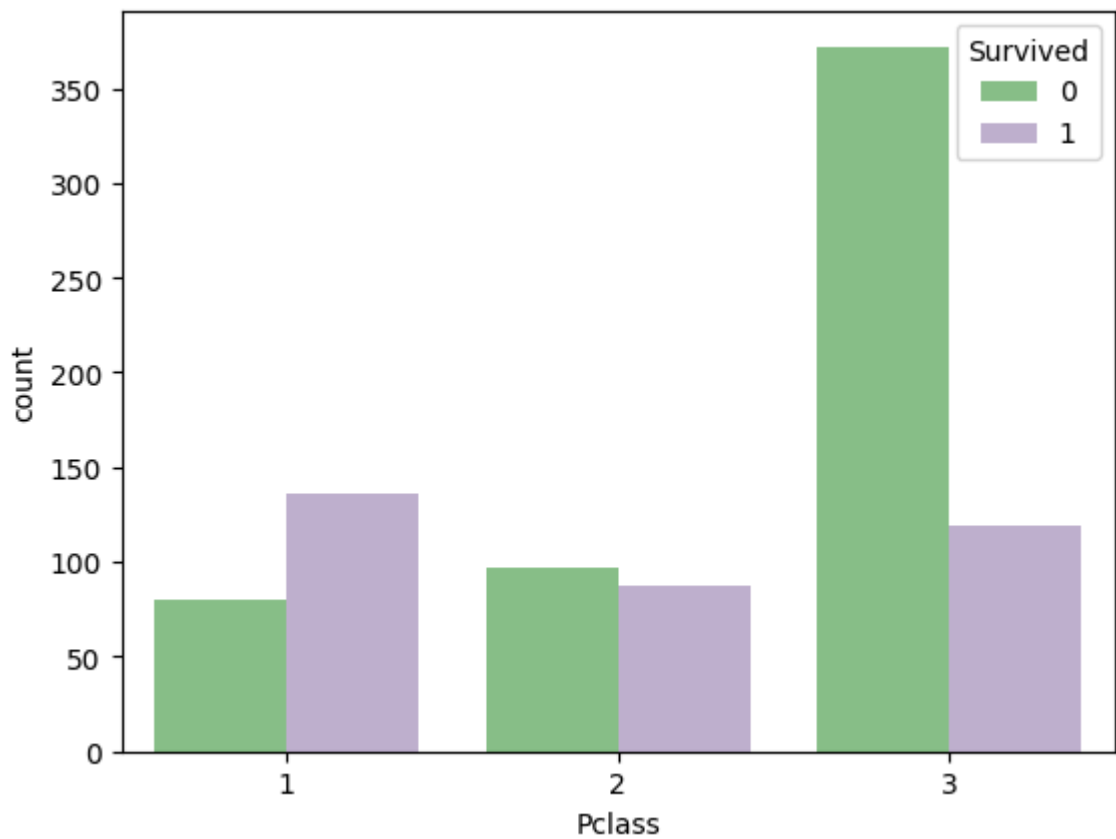
```
In [26]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 11 columns):
 #   Column      Non-Null Count  Dtype  
---  --
 0   PassengerId 891 non-null    int64  
 1   Survived    891 non-null    int64  
 2   Pclass      891 non-null    int64  
 3   Name        891 non-null    object  
 4   Sex         891 non-null    int64  
 5   Age         891 non-null    float64 
 6   SibSp       891 non-null    int64  
 7   Parch       891 non-null    int64  
 8   Ticket      891 non-null    object  
 9   Fare        891 non-null    float64 
10   Embarked    891 non-null    int64  
dtypes: float64(2), int64(7), object(2)
memory usage: 76.7+ KB
```

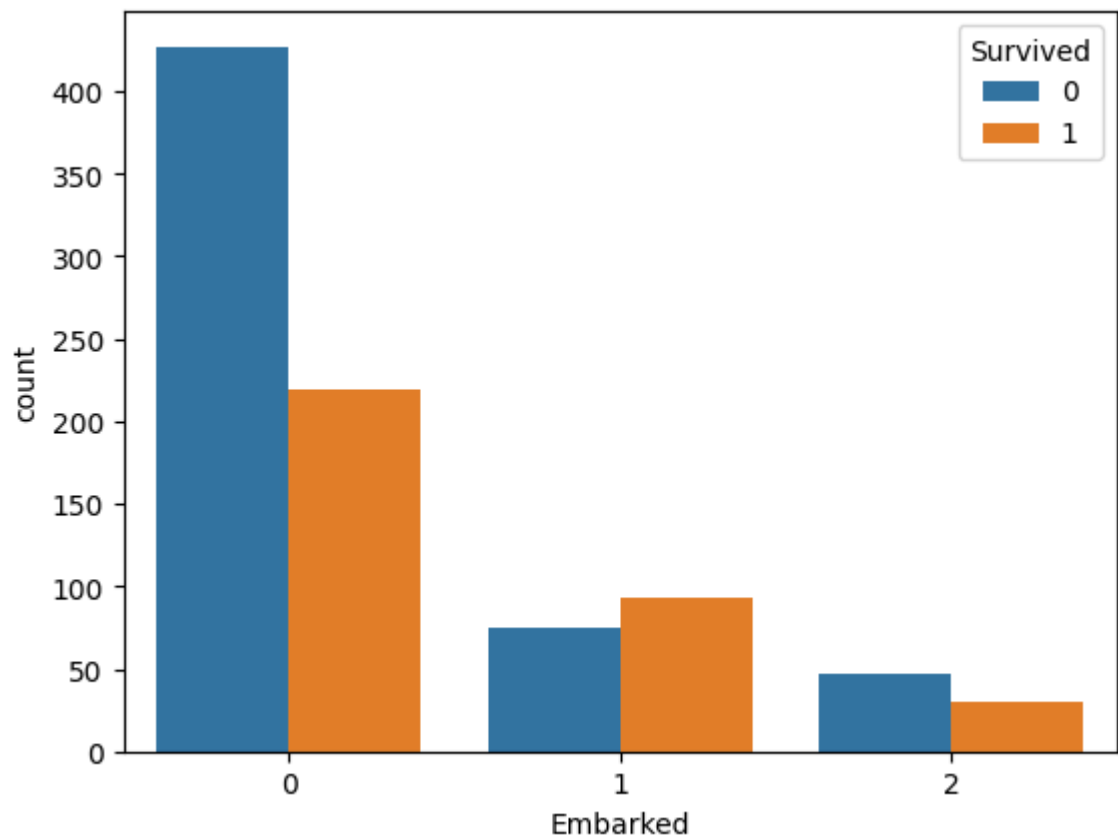
```
In [27]: sns.countplot(df, x="Survived")
plt.show()
```



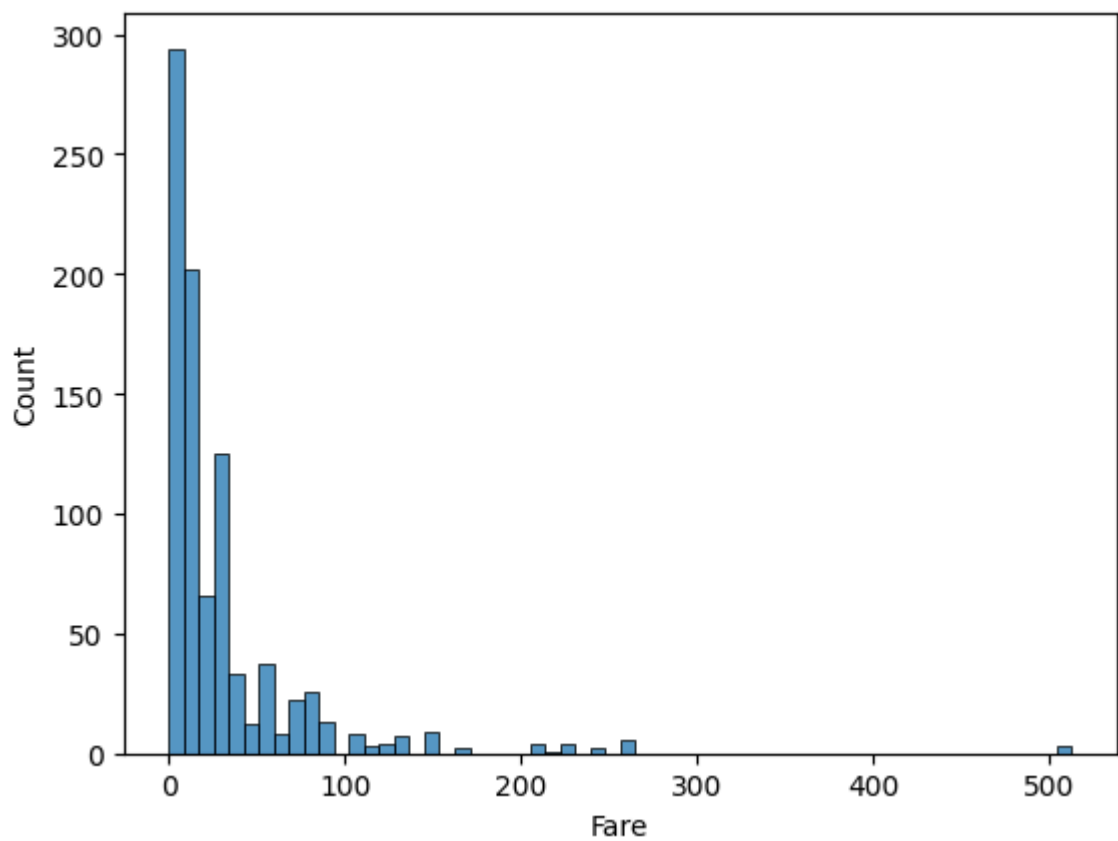
```
In [28]: sns.countplot(df,x="Pclass", hue="Survived",palette="Accent")  
plt.show()
```



```
In [29]: sns.countplot(df,x="Embarked",hue="Survived")  
plt.show()
```



```
In [30]: sns.histplot(df["Fare"])  
plt.show()
```



```
In [ ]:
```